

# Justin Lubin

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Curriculum Vitae, July 2026

## Research Vision

**Mission:** To enable scientists to write code with only domain expertise, not programming expertise.

**I co-design programming systems with scientists.** These systems empower scientists without a background in computing to write the code they need by themselves. To make new user interactions possible in these systems, I develop **programming language theory** informed by what I learn from my **deep embedding with domain experts** and my mixed methods human-computer interaction research.

**Primary area:** Programming languages   **Secondary areas:** Human-computer interaction, biology

## Education

**University of California, Berkeley** 2020–  
PhD in Computer Science  
Advisor: Sarah E. Chasins  
Major: Programming languages  
Minor: Bioinformatics

**University of Chicago** 2016–2020  
BS in Computer Science (Honors), BS in Mathematics, Minor in Music. *Summa cum laude*.  
Advisor: Ravi Chugh  
Honors Thesis: [Forging Smyth: The Impl. of Program Sketching with Live Bidirectional Evaluation](#)

## Research Internships

**Carnegie Mellon University** 2018  
Advisors: Jonathan Aldrich (CMU), Alex Potanin (Victoria University of Wellington)  
Project: [Approximating Polymorphic Effects with Capabilities](#)

## Peer-Reviewed Conference and Journal Publications

\* = equal contribution, † = research mentee

**Exploring the Translation-As-Explanation Folk Wisdom for Program Editing Tasks** VL/HCC '26  
Jacob Yim<sup>\*†</sup>, **Justin Lubin**<sup>\*</sup>, Kevin Ye<sup>†</sup>, Laila Walker<sup>†</sup>, Eric Rawn, and Sarah E. Chasins  
To appear in *IEEE Symposium on Visual Languages and Human Centric Computing* (2026).

**Navigating AND–OR Graph Modifications to Debug Failing Proof Search** PLDI '26  
**Justin Lubin**, Marlena Preigh<sup>†</sup>, Max Willsey, and Sarah E. Chasins  
In *Proceedings of the ACM on Programming Languages (PACMPL), Issue PLDI* (2026).

**Programming By Scaffolded Demonstration with Perpend** CHI '26  
Angela Bi<sup>\*</sup>, Eric Rawn<sup>\*</sup>, **Justin Lubin**, and Sarah E. Chasins  
In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (2026).

**Programmable epigenome editing by transient delivery of CRISPR epigenome editor ribonucleoproteins** Nat. Commun. '25

Da Xu\*, Swen Besselink\*, Gokul N. Ramadoss, Philip H. Dierks, **Justin P. Lubin**, Rithu K. Pattali, Jinna I. Brim, Anna E. Christenson, Peter J. Colias, Izaiah J. Ornelas, Sarah E. Chasins, Bruce R. Conklin, and James K. Nuñez

In *Nature Communications* (2025).

**Programming by Navigation**

PLDI '25

**Justin Lubin**, Parker Ziegler, and Sarah E. Chasins

In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2025).

(Selected for [MIT PL Review 2026](#).)

**Fast Direct Manipulation Programming with Patch-Reconciliation Correspondence**

PLDI '25

Parker Ziegler, **Justin Lubin**, and Sarah E. Chasins

In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2025).

**Equivalence by Canonicalization for Synthesis-Backed Refactoring**

PLDI '24

**Justin Lubin**, Jeremy Ferguson<sup>\*†</sup>, Kevin Ye<sup>\*†</sup>, Jacob Yim<sup>\*†</sup>, and Sarah E. Chasins

In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2024).

**Exploring the Learnability of Program Synthesizers by Novice Programmers**

UIST '22

Dhanya Jayagopal<sup>\*†</sup>, **Justin Lubin**<sup>\*</sup>, and Sarah E. Chasins

In *Proceedings of the ACM Symposium on User Interface Software and Technology* (2022).

**How Statically-Typed Functional Programmers Write Code**

OOPSLA '21

**Justin Lubin** and Sarah E. Chasins

In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue OOPSLA (2021).

**Program Sketching with Live Bidirectional Evaluation**

ICFP '20

**Justin Lubin**, Nick Collins, Cyrus Omar, and Ravi Chugh

In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue ICFP (2020).

**Sketch-n-Sketch: Output-Directed Programming for SVG**

UIST '19

Brian Hempel, **Justin Lubin**, and Ravi Chugh

In *Proceedings of the ACM Symposium on User Interface Software and Technology* (2019).

**Deuce: A Lightweight User Interface for Structured Editing**

ICSE '18

Brian Hempel, **Justin Lubin**, Grace Lu, and Ravi Chugh

In *Proceedings of the International Conference on Software Engineering* (2018).

## Preprints

<sup>\*</sup> = equal contribution

**Programmable CRISPRtune dissects the transcriptional repressive activity of MeCP2**

Jinna I. Brim\*, Izaiah J. Ornelas\*, Peter J. Colias, Nikita S. Divekar, Da Xu, **Justin P. Lubin**, Sophia I. Ferrel, Luis Galán Palma, Mitzi G. Hernández Zamora, Rithu K. Pattali, Jameel J. McDaniel, Sarah E. Chasins, and James K. Nuñez

## LEMONmethyl-seq: Targeted long-read DNA methylation profiling reveals dynamics of CRISPR epigenome editing and endogenous DNA methylation patterns

Anna E. Christenson, Nikita S. Divekar, **Justin P. Lubin**, Luis G. Palma, Peter J. Colias, Rithu K. Pattali, Da Xu, Akane Hubbard, Katie Lin, Ngan Phan, Bernardo Moreno, Sarah E. Chasins, S. John Liu, James K. Nuñez

## Transcriptional regulation of disease-relevant microglial activation programs

Amanda McQuade, Reet Mishra, Venus Hagan, Weiwei Liang, Peter J. Colias, Vincent Cele Castillo, Bianca Gonzalez, **Justin P. Lubin**, Verena Haage, Victoria Marshe, Masashi Fujita, Thomas Ta, Layla Gomes, Olivia Teter, Xia Han, Nathaniel Robichaud, Sarah E. Chasins, Jessica E. Rexach, Philip L. De Jager, James K. Nuñez, Martin Kampmann

## Workshop Papers

*\* = equal contribution, † = research mentee*

### Interactive Design Space Exploration for Sparse Accelerators

YArch '26

Evan Williams<sup>†</sup>, **Justin Lubin**, Rubens Lacouture, and Olivia Hsu  
Presented by Evan at *Young Architect Workshop*.

### Interactive Compiler Scheduling with Strong Guarantees

LATTE '26

Evan Williams<sup>†</sup>, **Justin Lubin**, Rubens Lacouture, and Olivia Hsu  
Presented by Evan at *Workshop on Languages, Tools, and Techniques for Accelerator Design*.

### Searching for Incidental Specifications

PLATEAU '23

Jeremy Ferguson<sup>\*†</sup>, Kevin Ye<sup>\*†</sup>, Jacob Yim<sup>\*†</sup>, and **Justin Lubin**  
In *Proceedings of the Workshop on Evaluation and Usability of Programming Languages and Tools*.

### Type-Directed Program Transformations for the Working Functional Programmer

PLATEAU '19

**Justin Lubin** and Ravi Chugh  
In *Proceedings of the Workshop on Evaluation and Usability of Programming Languages and Tools*.

## Selected Honors

### Research Recognition

MIT PL Review selection of *Programming by Navigation* 2026  
NSF Graduate Research Fellowship (GRFP) 2020–2025

### Mentoring and Teaching Awards

Undergraduate Research Mentoring Award (UC Berkeley) 2023  
Outstanding Graduate Student Instructor Award (UC Berkeley) 2022

### Student Research Competitions

3<sup>rd</sup> Place Graduate at CHI, *How Statically-Typed Functional Programmers Author Code* 2021  
1<sup>st</sup> Place Undergraduate at SPLASH, *Approximating Polymorphic Effects with Capabilities* 2018

### Special Recognition for Outstanding Review

CHI '23, UIST '25 (×2)

## Academic Honors

|                                              |           |
|----------------------------------------------|-----------|
| Enrico Fermi Scholar (University of Chicago) | 2020      |
| Harper Award (University of Chicago)         | 2020      |
| Student Marshal (University of Chicago)      | 2019–2020 |
| Dean's List (University of Chicago)          | 2016–2020 |

## Honor Societies

Phi Beta Kappa (2019–), Sigma Xi (2020–)

## Selected Presentations, Workshops, and Lectures

### Invited External Presentations

|                                                                                                                                    |      |
|------------------------------------------------------------------------------------------------------------------------------------|------|
| <a href="#">Pacific Programming Interfaces Confab (PPIC)</a> — Programming by Navigation with Honeybee                             | 2026 |
| <a href="#">MIT Programming Languages Review</a> — Programming by Navigation                                                       | 2026 |
| <a href="#">Stanford Software Research Lunch</a> — Programming by Navigation                                                       | 2025 |
| UCSD PL/HCI Graduate Seminar (Philip Guo) — Q&A on <i>Exploring the Learnability of Program Synthesizers by Novice Programmers</i> | 2025 |
| <a href="#">UCSC Languages, Systems, and Data Seminar</a> — Programming by Navigation                                              | 2024 |
| <a href="#">UPenn PLClub</a> — How Statically-Typed Functional Programmers Write Code                                              | 2022 |
| <a href="#">UCSD PL Seminar</a> — How Statically-Typed Functional Programmers Write Code                                           | 2021 |

### Workshops for Biologists

|                                                                                                |      |
|------------------------------------------------------------------------------------------------|------|
| <a href="#">Chan Zuckerberg Biohub</a> — Intro to Genomics Analysis using Honeybee             | 2026 |
| ◦ Over 40 attendees from Stanford, UCSF, UC Berkeley, and the Chan Zuckerberg Biohub           |      |
| <a href="#">Nuñez Lab</a> — From .FASTQs to .SVGs: How to Make a Bioinformatics Pipeline       | 2025 |
| <a href="#">Nuñez Lab</a> — Wrangling Flow Cytometry Data                                      | 2024 |
| <a href="#">Neurona Therapeutics</a> — Visualizing Data in Python with the Grammar of Graphics | 2023 |
| <a href="#">Neurona Therapeutics</a> — Getting Started with Python for Wet Lab Data Analysis   | 2023 |
| <a href="#">Nuñez Lab</a> — Visualizing Data in Python with the Grammar of Graphics            | 2023 |
| <a href="#">Nuñez Lab</a> — Analysis and Visualization of RNA-seq Data                         | 2023 |

### Guest Lectures (UC Berkeley)

|                                                                                                         |      |
|---------------------------------------------------------------------------------------------------------|------|
| <a href="#">Building User-Centered Programming Tools</a> — Statistics for the Working PL+HCI Researcher | 2026 |
| <a href="#">Building User-Centered Programming Tools</a> — User-Centered Design                         | 2023 |
| <a href="#">Programming Languages and Compilers</a> — Compiler Optimizations                            | 2021 |

## Professional Activities and External Service

### Dagstuhl Participation

|                                         |      |
|-----------------------------------------|------|
| <a href="#">Theories of Programming</a> | 2022 |
|-----------------------------------------|------|

### Organizing Committee

|                                                                         |             |
|-------------------------------------------------------------------------|-------------|
| Workshop on Evaluation and Usability of Programming Languages and Tools | PLATEAU '26 |
| Workshop on Evaluation and Usability of Programming Languages and Tools | PLATEAU '25 |
| Workshop on Evaluation and Usability of Programming Languages and Tools | PLATEAU '24 |
| ◦ Additional Responsibilities: Publicity Chair                          |             |
| Workshop on Evaluation and Usability of Programming Languages and Tools | PLATEAU '23 |

## Program Committee

|                                                                                  |             |
|----------------------------------------------------------------------------------|-------------|
| Workshop on Human Aspects of Types and Reasoning Assistants                      | HATRA '26   |
| Workshop on Live Programming                                                     | LIVE '26    |
| Workshop on Human Aspects of Types and Reasoning Assistants                      | HATRA '25   |
| Workshop on Live Programming                                                     | LIVE '25    |
| Workshop on Human Aspects of Types and Reasoning Assistants                      | HATRA '24   |
| Workshop on Live Programming                                                     | LIVE '24    |
| Symposium on New Ideas in Programming and Reflections on Software (Papers Track) | Onward! '24 |
| Workshop on Human Aspects of Types and Reasoning Assistants                      | HATRA '23   |
| Workshop on Live Programming                                                     | LIVE '23    |

## Reviewer

|                                                            |          |
|------------------------------------------------------------|----------|
| ACM Symposium on User Interface Software and Technology    | UIST '25 |
| ◦ Received Special Recognition for Outstanding Review (×2) |          |
| ACM Conference on Human Factors in Computing Systems       | CHI '25  |
| Journal of Functional Programming                          | JFP '24  |
| ACM Conference on Human Factors in Computing Systems       | CHI '23  |
| ◦ Received Special Recognition for Outstanding Review      |          |
| Formal Methods in System Design                            | FMSD '22 |
| ACM Symposium on User Interface Software and Technology    | UIST '21 |

## Student Volunteer

|                                                           |      |
|-----------------------------------------------------------|------|
| PL+HCI “Swimmer” School (Social Sessions Co-organizer)    | 2022 |
| Elm in the Spring Conference                              | 2019 |
| International Conference on Functional Programming (ICFP) | 2018 |

## Mentorship

*UG = undergrad*

### PhD Student Research Mentees

|                                                          |       |
|----------------------------------------------------------|-------|
| Marlena Preigh (advisor: Sarah E. Chasins @ UC Berkeley) | 2025– |
| ◦ Co-authorship at PLDI '26                              |       |
| Evan Williams (advisor: Olivia Hsu @ CMU)                | 2025– |
| ◦ Co-mentored with Rubens Lacouture @ Stanford           |       |
| ◦ First authorship at LATTE '26 and YArch '26            |       |

### MS Thesis Research Mentees

|                                                                                                                   |           |
|-------------------------------------------------------------------------------------------------------------------|-----------|
| Jacob Yim (UG, MS @ UC Berkeley → PhD @ UCSD)                                                                     | 2022–2025 |
| ◦ <i>MS Thesis</i> : <a href="#">Translations Alone Do Not Help Programmers Work With Unfamiliar Abstractions</a> |           |
| ◦ Co-first authorship at VL/HCC '26                                                                               |           |
| ◦ Co-authorship at PLDI '24                                                                                       |           |
| ◦ Co-first authorship at PLATEAU '23                                                                              |           |
| Jeremy Ferguson (UG, MS @ UC Berkeley → Software Engineer @ Voleon)                                               | 2022–2024 |
| ◦ <i>MS Thesis</i> : <a href="#">Eliciting Domain Expertise Reduces Examples Needed for Program Synthesis</a>     |           |
| ◦ Co-authorship at PLDI '24                                                                                       |           |
| ◦ 2 <sup>nd</sup> place UG at POPL SRC '23, <i>Synthesizing Vectorized Code via Verified Lifting</i>              |           |
| ◦ Co-first authorship at PLATEAU '23                                                                              |           |

Dhanya Jayagopal (UG, MS @ UC Berkeley → Software Engineer @ UdeMy) 2020–2022  
 ◦ MS Thesis: [Study of Program Synthesizers & Novice Programmers](#)  
 ◦ Co-first authorship at UIST '22

#### **Additional Research Mentees**

Sam Poder (UG @ UC Berkeley) 2025–2026  
 Laila Walker (UG @ UC Berkeley → PhD @ UW) 2024–2025  
 ◦ Co-mentored with MS Thesis Research Mentee Jacob Yim  
 ◦ Co-authorship at VL/HCC '26  
 Kevin Ye (UG, Gap-year Researcher @ UC Berkeley → MS @ Simon Fraser University) 2022–2024  
 ◦ Co-authorship at VL/HCC '26  
 ◦ Co-authorship at PLDI '24  
 ◦ Co-first authorship at PLATEAU '23  
 Fayaz Shaik (UG @ Ohlone College & [TTE REU](#) Mentee → UG @ UCSD) 2021–2022  
 ◦ Poster at SACNAS NDiSTEM '21, *User-Centered Data Population of Knowledge Graphs*

#### **SIGPLAN-M Mentees**

Carolina Carreira (PhD @ CMU Portugal) 2022–  
 Adharsh Kamath (Research Fellow @ MSR India, PhD @ UIUC) 2022–

#### **Additional Mentees**

Kasozi Vincent (MS @ University of Chlef) 2022–2023

## **Teaching**

### **University of California, Berkeley**

CS 164: [Programming Languages and Compilers](#) (TA) Fall 2022  
 CS 164: [Programming Languages and Compilers](#) (Head TA) Fall 2021  
 ◦ Received Outstanding Graduate Student Instructor Award

### **University of Chicago**

CMSC 22300: [Functional Programming](#) (TA) Spring 2020  
 CMSC 16100: [Honors Introduction to Programming I](#) (Grader) Fall 2018

## **Internal Service**

### **UC Berkeley [Programming Systems \(PS\) Research Group](#) and [EECS Department](#)**

EECS Faculty Hiring Committee, Student Member 2025, 2026  
 EECS Visit Days Peer Advisor '21, '22, '23, '25, '26  
 PS Visit Days Coordinator 2022  
 PS Seminar Organizer 2021–2022  
 PS Graduate Admissions Representative 2021

## **Leadership, Diversity, and Community Service**

[Incarceration to College Program](#) — Tutor 2025–  
 Tutoring incarcerated youth in mathematics to prepare for post-secondary education.

- Berkeley Underground Scholars — Graduate Tutor** 2023–  
Tutoring formerly-incarcerated/systems-impacted students in computer science and mathematics.
- EECS Peers — Co-organizer** 2023–  
Co-leading a graduate student peer support and mentorship organization. (*Mentor since 2022.*)
- Equal Access to Application Assistance Program — Reviewer** 2023–  
Reviewed PhD applications as part of a student-run initiative aiming to ensure that all PhD applicants to UC Berkeley have access to guidance on the application process.
- SIGPLAN-M Long-Term Mentorship Program — Mentor** 2022–  
Serving as a mentor for the SIGPLAN-M long-term programming languages mentorship program.
- Association of Women in EE&CS (AWE) — Grad School Advisor** 2023  
Hosted series of office hours to advise women and nonbinary undergraduates interested in grad school.
- WiCSE Girl Scout Engineering Day — Activity Designer and Leader** 2023  
Designed and led educational activities for the Girl Scouts of Northern California to earn their programming and robotics badge.
- Women in Computer Science and Electrical Engineering (WiCSE) — Grad Mentor** 2022  
Served as a mentor for the WiCSE Grad Mentorship Program, where elder PhD students get paired up with first- and second-year PhD students for general mentorship.
- #ShutdownPL — Co-organizer** 2020–2022  
Coordinated logistics and managed funding for three keynote speakers at dedicated anti-racist events at ICFP and POPL.
- Transfer-to-Excellence Summer Research Program (TTE REU) — Mentor** 2021  
Served as a research mentor for the TTE REU, a research experience for community college students seeking to transfer to four-year universities.
- compileHer Tech Capstone — Activity Leader** 2019  
Led computer science educational activities for middle school girls in Chicago.